

Restricting and sorting data

1) Display the **ENAME, SAL, COMM** from the **EMP** table who earn commission and sort the records in **descending order of Salary and Commission**. Use **column's numeric position** in the ORDER BY clause.

```
select ename,sal,comm from emp where comm is not null order by 2 desc,3 desc;
```

Output:-

```
13 rows selected.
SQL> select ename,sal,comm from emp where comm is not null order by 2 desc,3 desc;

ENAME            SAL      COMM
-----
ALLEN             1600      300
TURNER            1500       0
MARTIN            1250     1400
WARD              1250      500

SQL> |
```

2) The HR department needs a query to display **all unique job codes** from the **EMP** table.

```
select distinct(job) from EMP;
```

Output:-

```
SQL> select distinct(job) from EMP;

JOB
-----
ANALYST
SALESMAN
CLERK
MANAGER
PRESIDENT

SQL> |
```

3) The HR department wants more descriptive column headings for its report on employees. Name the column headings **Emp #, Employee, Job, and Hire Date**, respectively, by giving **Column Alias**.

```
select EMPNO AS "EMP #",ENAME AS EMPLOYEE_NAME,JOB AS Job,MGR AS
MANAGER,HIREDATE AS "HIRE DATE",SAL AS SALARY,COMM AS COMMISSION,DEPTNO AS
DEPT_NO from emp;
```

Output:-

SQL Plus					
15 rows selected.					
SQL> select empno as "EMP #",ENAME AS EMPLOYEE_NAME,JOB AS Job,MGR AS MANAGER,HIREDATE AS "HIRE DATE",SAL AS SALARY,COMM AS COMMISSION,DEPTNO AS DEPT_NO from emp;					
EMP #	EMPLOYEE_NAME	JOB	MANAGER	HIRE DATE	

SALARY	COMMISSION	DEPT_NO			
-----	-----	-----			
100	SCOTT	ANALYST		106 19-APR-81	
3900		20			
103	TURNER	SALESMAN		105 08-SEP-81	
1500	0	30			
104	SMITH	CLERK		106 17-DEC-80	
800		20			

EMP #	EMPLOYEE_NAME	JOB	MANAGER	HIRE DATE	

SALARY	COMMISSION	DEPT_NO			
-----	-----	-----			
105	BLAKE	MANAGER		101 01-MAY-81	
2850		30			
106	JONES	MANAGER		101 02-APR-81	
2975		20			
107	ALLEN	SALESMAN		105 20-FEB-81	
1600	300	30			

EMP #	EMPLOYEE_NAME	JOB	MANAGER	HIRE DATE	

SALARY	COMMISSION	DEPT_NO			
-----	-----	-----			
108	WARD	SALESMAN		105 22-FEB-81	
1250	500	30			
109	MARTIN	SALESMAN		105 28-SEP-81	
1250	1400	30			
110	JAMES	CLERK		105 03-DEC-81	
950		30			

EMP #	EMPLOYEE_NAME	JOB	MANAGER	HIRE DATE	

SALARY	COMMISSION	DEPT_NO			
-----	-----	-----			
111	MILLER	CLERK		101 23-JAN-82	
1300		10			
112	CLARK	MANAGER		101 09-JUN-81	
2450		10			
113	ADAMS	CLERK		102 23-MAY-87	
1100		20			

EMP #	EMPLOYEE_NAME	JOB	MANAGER	HIRE DATE	

SALARY	COMMISSION	DEPT_NO			
-----	-----	-----			
114	SARAH	CLERK		112 16-AUG-81	
2100		10			
115	RAMESH	ANALYST		112 10-OCT-81	
2000		20			
101	KING	PRESIDENT		17-NOV-81	
5000		10			

15 rows selected.					
SQL>					

4) The HR department has requested a report of all employees and their job IDs. Display the **last name concatenated with the job ID** (separated by a comma and a space) and name the column **Employee and Title** by giving **Column Alias**.

```
SELECT ename || ', ' || job AS "Employee and Title" FROM emp;
```

Output:-

SQL> SELECT ename ', ' job AS "Employee and Title" FROM emp;
Employee and Title

SCOTT, ANALYST
TURNER, SALESMAN
SMITH, CLERK
BLAKE, MANAGER
JONES, MANAGER
ALLEN, SALESMAN
WARD, SALESMAN
MARTIN, SALESMAN
JAMES, CLERK
MILLER, CLERK
CLARK, MANAGER
Employee and Title

ADAMS, CLERK
SARAH, CLERK
RAMESH, ANALYST
KING, PRESIDENT
15 rows selected.
SQL>

5) To familiarize yourself with the data in the **EMP** table, create a query to display all the data from that table. Separate each column output by a comma. **ENAME, JOB, HIREDATE, MGR**. Name the column title **THE_OUTPUT**.

```
SELECT empno || ',' || ename || ',' || job || ',' || mgr || ',' || hiredate || ',' || sal || ',' || comm || ',' || deptno  
AS "THE_OUTPUT" FROM emp;
```

Output:-

```
SQL> SELECT empno || ',' || ename || ',' || job || ',' || mgr || ',' || hiredate || ',' || sal || ',' || comm || ',' || deptno AS "THE_OUTPUT" FROM emp;  
  
THE_OUTPUT  
-----  
102, SCOTT, ANALYST, 106, 19-APR-81, 3000, , 20  
103, TURNER, SALESMAN, 105, 08-SEP-81, 1500, 0, 30  
104, SMITH, CLERK, 106, 17-DEC-80, 800, , 20  
105, BLAKE, MANAGER, 101, 01-MAY-81, 2850, , 30  
106, JONES, MANAGER, 101, 02-APR-81, 2975, , 20  
107, ALLEN, SALESMAN, 105, 20-FEB-81, 1600, 300, 30  
108, WARD, SALESMAN, 105, 22-FEB-81, 1250, 500, 30  
109, MARTIN, SALESMAN, 105, 28-SEP-81, 1250, 1400, 30  
110, JAMES, CLERK, 105, 03-DEC-81, 950, , 30  
111, MILLER, CLERK, 101, 23-JAN-82, 1300, , 10  
112, CLARK, MANAGER, 101, 09-JUN-81, 2450, , 10  
  
THE_OUTPUT  
-----  
113, ADAMS, CLERK, 102, 23-MAY-87, 1100, , 20  
114, SARAH, CLERK, 112, 15-AUG-81, 2100, , 10  
115, RAMESH, ANALYST, 112, 10-OCT-81, 2600, , 20  
101, KING, PRESIDENT, , 17-NOV-81, 5000, , 10  
  
15 rows selected.
```

6) Create a report to display the **ENAME, JOB, and HIREDATE** for the employees whose name is **SCOTT** or **TURNER**. Order the query in **ascending order by HIREDATE**.

```
SELECT ENAME, JOB, HIREDATE  
FROM EMP  
WHERE ENAME IN ('SCOTT', 'TURNER')  
ORDER BY HIREDATE ASC;
```

Output:-

```
SQL> SELECT ENAME, JOB, HIREDATE  
2 FROM EMP  
3 WHERE ENAME IN ('SCOTT', 'TURNER')  
4 ORDER BY HIREDATE ASC;  
  
ENAME          JOB          HIREDATE  
-----  
SCOTT          ANALYST     19-APR-81  
TURNER         SALESMAN    08-SEP-81  
  
SQL>
```

7) Display the **ENAME** and **DEPTNO** of all employees in departments **20** or **30** in **ascending alphabetical order by ENAME**.

```
SELECT ENAME, DEPTNO  
FROM EMP  
WHERE DEPTNO IN (20, 30)  
ORDER BY ENAME ASC;
```

Output:-

```
SQL> SELECT ENAME, DEPTNO
2 FROM EMP
3 WHERE DEPTNO IN (20, 30)
4 ORDER BY ENAME ASC;
```

ENAME	DEPTNO
ADAMS	20
ALLEN	30
BLAKE	30
JAMES	30
JONES	20
MARTIN	30
RAMESH	20
SCOTT	20
SMITH	20
TURNER	30
WARD	30

11 rows selected.

8) Modify the previous query to display the **last name and salary** of employees who earn **between 2000 and 3000** and are in department **20 or 30**. Label the columns **Employee** and **Monthly Salary**, respectively, by giving **Column Alias**.

```
select ename as Employee,sal as Monthly_salary from emp
where deptno in(20,30) and sal between 2000 and 3000;
```

Output:-

```
SQL> select ename as Employee,sal as Monthly_salary from emp
2 where deptno in(20,30) and sal between 2000 and 3000;
```

EMPLOYEE	MONTHLY_SALARY
SCOTT	3000
BLAKE	2850
JONES	2975
RAMESH	2600

SQL>

9) The HR department needs a report that displays the **last name and hire date** for all employees who were **hired in 1981**.

```
SELECT ENAME, HIREDATE
FROM EMP
WHERE EXTRACT(YEAR FROM HIREDATE) = 1981;
```

Output:-

```
SQL> SELECT ENAME, HIREDATE
2 FROM EMP
3 WHERE EXTRACT(YEAR FROM HIREDATE) = 1981;

ENAME                HIREDATE
-----
SCOTT                19-APR-81
TURNER              08-SEP-81
BLAKE               01-MAY-81
JONES              02-APR-81
ALLEN              20-FEB-81
WARD              22-FEB-81
MARTIN            28-SEP-81
JAMES             03-DEC-81
CLARK             09-JUN-81
SARAH            15-AUG-81
RAMESH           10-OCT-81

ENAME                HIREDATE
-----
KING               17-NOV-81

12 rows selected.

SQL>
```

10) Display the **ENAME** and **SAL** of employees who earn **more than an amount the user specifies after a prompt**.

```
select ename,sal from emp where sal>&sal;
```

Output:-

```
SQL> select ename,sal from emp where sal>&sal;
Enter value for sal: 1200
old  1: select ename,sal from emp where sal>&sal
new  1: select ename,sal from emp where sal>1200

ENAME                SAL
-----
SCOTT                3000
TURNER              1500
BLAKE               2850
JONES              2975
ALLEN              1600
WARD              1250
MARTIN            1250
MILLER             1300
CLARK             2450
SARAH             2100
RAMESH            2600

ENAME                SAL
-----
KING               5000

12 rows selected.

SQL>
```

11) Create a report to display the **last name and job title** of all employees who **do not have a manager**.

```
select ename,job from emp where mgr is null;
```

Output:-

```
SQL> select ename,job from emp where mgr is null;

ENAME                JOB
-----
KING                 PRESIDENT

SQL> |
```

12) Create a query that **prompts the user for Manager ID** and generates **EMPNO, ENAME, SAL, DEPTNO**. The user should have the ability to **sort the records on a selected column**.

```
SELECT EMPNO, ENAME, SAL, DEPTNO
FROM EMP
WHERE MGR = &MGR
ORDER BY &COLUMN;
```

Output:-

```
SQL> SELECT EMPNO, ENAME, SAL, DEPTNO
2  FROM EMP
3  WHERE MGR = &MGR
4  ORDER BY &COLUMN;
Enter value for mgr: 101
old  3: WHERE MGR = &MGR
new  3: WHERE MGR = 101
Enter value for column: 3
old  4: ORDER BY &COLUMN
new  4: ORDER BY 3

EMPNO ENAME                SAL    DEPTNO
-----
111 MILLER                1300     10
112 CLARK                 2450     10
105 BLAKE                 2850     30
106 JONES                 2975     20

SQL> |
```

13) The HR department wants to run reports based on a manager. Create a query that **prompts the user for a MGR** and generates the **EMPNO, ENAME, SALARY, and DEPARTMENT** for that manager's employees. The HR department wants the ability to **sort the report on a selected column**.

```
SELECT EMPNO, ENAME, SAL, DEPTNO
FROM EMP
WHERE MGR = &MGR
ORDER BY &COLUMN;
```

OutPut:-

```
SQL> SELECT EMPNO, ENAME, SAL, DEPTNO
  2  FROM EMP
  3  WHERE MGR = &MGR
  4  ORDER BY &COLUMN;
Enter value for mgr: 102
old  3: WHERE MGR = &MGR
new  3: WHERE MGR = 102
Enter value for column: 2
old  4: ORDER BY &COLUMN
new  4: ORDER BY 2

      EMPNO ENAME                SAL      DEPTNO
-----
      113 ADAMS                1100         20

SQL> |
```

14) Display all employee **last names** in which the **third letter** of the name is **A**.

```
SELECT ENAME
FROM EMP
WHERE ENAME LIKE '___A%';
```

Output:-

```
SQL> SELECT ENAME
  2  FROM EMP
  3  WHERE ENAME LIKE '___A%';

ENAME
-----
BLAKE
CLARK
ADAMS

SQL> |
```

15) Display the **last name** of all employees who have **both an A and an S** in their name.

```
SELECT ENAME
FROM EMP
WHERE ENAME LIKE '%A%'
AND ENAME LIKE '%S%';
```

Output:-

```
SQL> SELECT ENAME
  2  FROM EMP
  3  WHERE ENAME LIKE '__A%';

ENAME
-----
BLAKE
CLARK
ADAMS

SQL> SELECT ENAME
  2  FROM EMP
  3  WHERE ENAME LIKE '%A%'
  4  AND ENAME LIKE '%S%';

ENAME
-----
JAMES
ADAMS
SARAH
RAMESH

SQL> |
```

16) Display the **ENAME**, **JOB**, **SAL** for all employees whose job is **CLERK** and whose salary is **800**, **950**, or **1300**.

```
SELECT ENAME, JOB, SAL
FROM EMP
WHERE JOB = 'CLERK'
AND SAL IN (800, 950, 1300);
```

Output:-

```
SQL> SELECT ENAME, JOB, SAL
  2  FROM EMP
  3  WHERE JOB = 'CLERK'
  4  AND SAL IN (800, 950, 1300);

ENAME                JOB                SAL
-----
SMITH                CLERK                800
JAMES                CLERK                950
MILLER               CLERK                1300

SQL>
```